CAPSTONE PROJECT REPORT

Procure-to-Pay (P2P) Cycle
with Procurement Analytics

A Study of SAP MM Procurement Process with Data-Driven Insights

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1. Introduction

The availability of a steady supply of raw materials and supplies is critical to every manufacturing organization of any size. Proper handling of the whole procurement process is the difference between smooth operations and having a constant firefighting situation with either stock-outs, production delays, or high overheads.

In essence, procurement involves assessing the requirements of the company, sourcing for the required goods, issuing of an official purchase request, receipt of the ordered items, verification of the order, and payment to the vendor. While simple on paper, the procurement process affects many departments in an organization such as stores, finance, and management, to mention but a few, and entails voluminous amounts of data collection at every stage.

It is at this point that SAP ERP becomes indispensable. Systems Applications and Products (SAP) is the world's leading ERP software used widely across industries to handle all business operations within an organization in an integrated fashion. The SAP MM manages the entire procurement cycle from purchase order through payment in a systematic fashion leaving behind a trail of information at every step.

The Procure-to-Pay (P2P) Process forms an integral part of SAP MM. This process pertains to the entire chain of events starting from identifying the need for materials within the organization to actually making payments to the supplier of the materials. Every step involved in this process has been assigned a particular SAP transaction code.

In this project, we will recreate a practical application of the P2P process through ABC Manufacturing Ltd., a medium-sized steel manufacturing firm. However, our goal here goes beyond simply understanding the process. In addition, we will be analyzing the procurement data collected by the organization with the aim of deriving actionable business intelligence.

> **Note on Simulation:** Due to limited access to a live SAP environment, all SAP transactions referenced in this report are conceptually simulated based on the standard SAP MM Procure-to-Pay process flow. The process steps, transaction codes, document flow, and business logic described here accurately reflect real-world SAP MM functionality as documented in standard SAP training and process guidelines.

2. Objective

The project has been designed to achieve three main objectives:

- To emulate the SAP Procure to Pay process as it is supposed to run in an actual manufacturing company, where every business task can be mapped to its corresponding SAP code.
- To collect procurement data generated by more than one purchase order and vendor, and then analyze it to make sense out of the expenditure made by the business.
- To evaluate the performance of the vendors in the procurement process, discover any trend in monthly procurement activities, and propose some useful changes to save costs or mitigate risks.

With this exercise, the goal is to connect the dots between the theory of SAP and its practice in the real business world.

3. Company Overview

In this exercise, a fictional firm named **ABC Manufacturing Limited** has been chosen.

Attribute	Details
Company Name	ABC Manufacturing Ltd
Industry	Steel and Metal Products Manufacturing
Core Activity	Procures raw materials (Steel Rod, Copper Wire, Iron Sheet) and converts them into finished goods
SAP Module Used	SAP MM (Materials Management)
ERP System	SAP ECC 6.0 (simulated environment)
Key Vendors	ABC Suppliers, XYZ Metals, PQR Traders

The organization obtains its raw materials from three vendors, who are approved suppliers to the company, and procures materials every month. All the material procurement activities are performed using the SAP MM Procure-to-Pay process.

4. Process Architecture

The P2P (Procure to Pay) process for ABC Manufacturing Ltd follows a linear approach. Every step must occur in sequence; you cannot post a Goods Receipt unless there is an associated Purchase Order, and you cannot post a payment without first verifying the associated invoice.

This strict sequence of events within SAP enforces a clean audit trail, along with the following general high level steps of the P2P process:

- Demand – The material requirement is determined by a department in the factory, and a demand document is raised.
- Purchase Requisition (PR) – The demand is then documented in SAP under the Purchase Requisition as a request for purchase.
- Purchase Order (PO) — The purchasing department converts the PR into a Purchase Order and sends it to the selected vendor.
- Goods Receipt (GR) – On receipt of the material, the GR process is performed by the warehousing department, and the document is uploaded to SAP.
- Invoice Verification – The vendor issues an invoice, which is checked for discrepancies against the PO and GR documents in SAP.
- Payment – Once the invoices are processed without any discrepancies, the payment is effected to the vendor from the finance module.

The flow diagram below provides a visual representation of this architecture:

Figure 1: P2P Process Flow Diagram — ABC Manufacturing Ltd

5. Steps in SAP P2P Process

Each step in the process of purchase-to-payment (P2P) in SAP is carried out by use of a particular transaction code. Hereunder is a step-by-step description of what each step entails in SAP ECC:

5.1 Purchase Requisition — ME51N

This is the first step in the process of purchase to payment. In case there is any shortage of raw materials within any production line within the organization, a PR will be raised using transaction ME51N.

The PR is an internal document that will be generated for purposes of communicating needs within the organization. The creator of the PR will indicate the item number, the quantity required, the date of delivery, and the respective plant.

5.2 Purchase Order — ME21N

After the PR approval, a Purchase Order (PO) is prepared by the procurement officer via transaction code ME21N. The PO is a legal document that includes information about the purchase such as what is being purchased, the quantity, price and the deadline.

The transaction is used to make reference to the PR while creating the PO for complete transparency. The vendor details, payment terms and delivery details are also recorded at this point. The vendors maintained by ABC Manufacturing Ltd include ABC Suppliers, XYZ Metals and PQR Traders from whom the best-suited one is chosen based on price and availability.

5.3 Goods Receipt — MIGO

Upon receiving the material, the store keeper creates a Goods Receipt (GR) on the PO through transaction MIGO. As soon as the GR is created, the inventory record is updated along with posting a journal entry regarding the value of goods received.

The quantity is compared with that purchased and any under-delivery or rejection of goods is recorded. The GR forms the basis for the following process of Invoice Processing.

5.4 Invoice Verification — MIRO

Once the items have been received from the supplier, the supplier will issue an invoice for the payment process. The process will then be handled by transaction MIRO

(Logistics Invoice Verification). SAP does the three way matching of PO, GR, and invoice.

Where there are inconsistencies such as where the supplier has issued an invoice for excess units than those received by the business, SAP sets a tolerance level or rejects the invoice entirely.

5.5 Payment — F-53

After the invoice has been checked and approved, the next stage is making payments. Using transaction F-53, the accounting team posts the debit entry to the bank account and credits the vendor's open item account.

It concludes one cycle in the P2P process. The documents generated at each stage (PR, PO, GR, Invoice, Payment) can be traced from start to finish in SAP.

Quick Reference: SAP P2P Transaction Codes

To quickly refer back to, I have listed below the summary of all five transaction codes used during the P2P process:

Step	Process Name	T-Code	Key Action
1	Purchase Requisition	**ME51N**	Raise internal material request
2	Purchase Order	**ME21N**	Send order to approved vendor
3	Goods Receipt	**MIGO**	Record materials received
4	Invoice Verification	**MIRO**	3-way match and bill clearance
5	Vendor Payment	**F-53**	Release outgoing payment

There are five transactions which form one document each in SAP, and all five documents are interrelated. Therefore, in case any auditor wants to trace a transaction from start to finish, then he can do so without difficulty.

6. Data Model

In order to evaluate the performance of the procurement function, a simple yet typical database was created using the purchase orders issued by ABC Manufacturing Ltd for a period of four months (January through April). The following columns are included in the database:

Field	Type	Description
Vendor	Text	Name of the supplying vendor
Material	Text	Type of raw material procured
Quantity	Numeric	Number of units ordered
Unit Price	Numeric (₹)	Price per unit charged by the vendor
Total Amount	Calculated (₹)	Derived using: Total = Quantity × Unit Price
Month	Text	Month in which the purchase order was raised

The central formula in the database is as follows:

Total Amount = Quantity * Unit Price

It is embedded in the excel spreadsheet as an excel formula (=C*D), making it possible to calculate dynamically and automatically each time an input value changes. The database also consists of a summary sheet where the SUMIF and COUNTIF formulas were applied for each vendor per month – this is the basis for the graphs.

7. Data Analysis

The procurement data was analyzed from two angles: vendor-wise performance and monthly spending trends. This gives management a clear picture of where the money is going and how demand fluctuates across the year.

7.1 Vendor-wise Analysis

The bar chart below shows the total procurement spend per vendor across all four months:

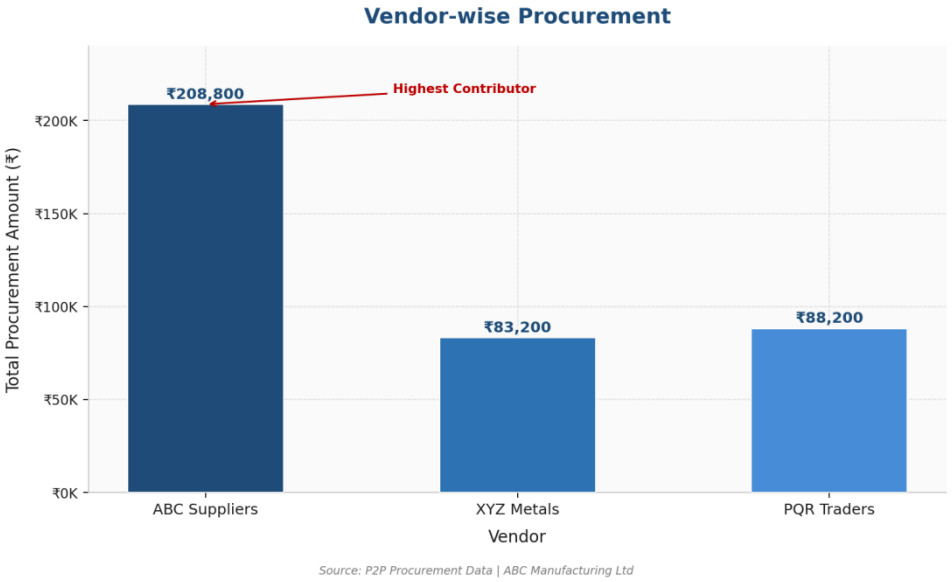

Figure 2: Vendor-wise Procurement — Total Amount (₹)

ABC Suppliers tops the chart by a clear margin, which makes sense given that they supply two different materials (Steel Rod and Iron Sheet) across all four months. This is partly because they supply across multiple material categories (Steel Rod and Iron Sheet) and across all four months. XYZ Metals and PQR Traders have a more limited and concentrated supply relationship.

From a supply chain perspective, having one vendor dominate the procurement mix is a risk. If ABC Suppliers were to face a disruption — production issues, pricing hike, or logistics problems — it could have a disproportionate impact on ABC Manufacturing Ltd's operations.

7.2 Monthly Procurement Trend

The following graph shows the trend in monthly procurement expenditure:

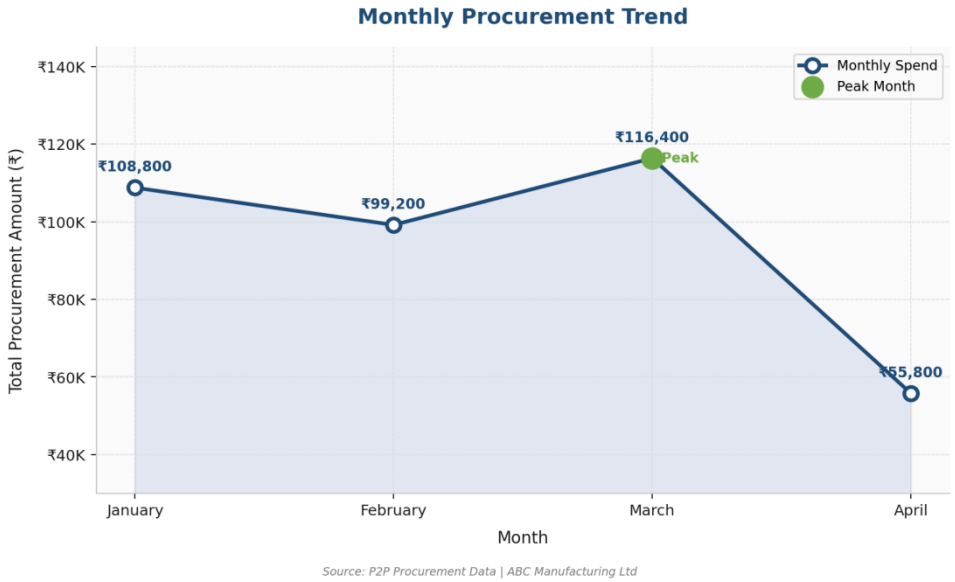

Figure 3: Monthly Procurement Trend (₹)

March saw the highest level of procurement expenditure that can be attributed to increased production objectives or stockpiling in anticipation of the year end. The levels of procurement expenditure remained stable during the first two months, while the last two months indicate a significant decrease in the volume – a result of seasonal demand downturn or reduction in inventory before ordering again.

7.3 Cost Analysis

When analyzing the cost structure of purchasing orders from different vendors using unit prices, there are certain observations to make. ABC Suppliers charge ₹490-500 per unit compared to ₹520 charged by XYZ Metals and ₹510 charged by PQR Traders for Steel Rods. Thus, the most attractive cost structure was offered by ABC Suppliers, which justified their high share of purchases.

PQR Traders charge ₹820 compared to ₹800 from XYZ Metals per unit for Copper Wire. For the Iron Sheets, ABC Suppliers charge ₹640-650. The cost structure per unit is minimal, but can be compounded significantly when analyzed on larger volumes. Regular price comparisons and negotiations should help in cutting additional expenses.

8. Business Insights

The data brings up three things worth paying attention to from a business standpoint:

Vendor Dependency Risk

ABC Suppliers handles the majority of ABC Manufacturing Ltd's procurement. That works fine as long as the relationship runs smoothly — but it is a single point of failure. A multi-vendor strategy — where at least two vendors are capable of supplying each material — would reduce this risk. Qualifying additional vendors for Steel Rod and Iron Sheet should be a priority.

Cost Optimization Opportunity

Price variation between vendors for the same material shows there is room for negotiation. Running competitive bidding or rate contracts for high-volume items like Steel Rod could yield meaningful savings. Even a 2-3% reduction in unit price on high-volume orders translates to thousands of rupees saved over a quarter.

Demand Fluctuation

The sharp drop in procurement in April suggests that demand is not linear throughout the year. This has implications for cash flow planning, inventory management, and vendor capacity planning. Developing a rolling procurement forecast — even a simple three-month look-ahead — would allow the purchasing team to plan better and potentially secure better pricing through advance commitments.

9. Results

Going through this simulation end-to-end made a few things clear that are easy to miss when you just read about SAP in theory:

- The P2P process in SAP ensures that every procurement activity is traceable — from the initial department request to the vendor payment — eliminating ambiguity and reducing the risk of unauthorized purchases.
- Once the procurement data is organized properly in a spreadsheet — with the right formulas pulling totals by vendor and month — things that are not obvious from raw records start becoming clear. Vendor concentration, spending spikes, price gaps between suppliers — all of it becomes visible.
- The three-way match capability in SAP (PO, GR, Invoice) is a powerful control mechanism that prevents overpayments and ensures the company pays only for what it actually receives and at the agreed price.
- A lot of procurement decisions in smaller organizations get made out of habit — the same vendor gets used simply because they were used before. Having the data in front of you gives the purchasing team a real reason to negotiate or explore alternatives.

10. Future Scope

The P2P process presented here is the essence of the process. There is ample scope to extend the process further, and some of those extensions are as follows:

- Migration to SAP S/4HANA: Large enterprises are gradually migrating from SAP ECC to SAP S/4HANA. While the core process flow remains more or less the same, there is immense improvement in terms of speed and reporting due to the in-memory database provided by SAP HANA. Mapping the exact process flow in S/4HANA will be an excellent continuation of the project.
- Power BI integration: The graphs within the Excel sheet are static. It means that they represent the data as it was at the moment of creation. However, if the same dataset is integrated into Power BI, it will give the procurement department more flexibility to analyze it in any way required, for example, by vendors, materials, month, or plants, and do it dynamically as new data appears.
- Demand forecasting: Once sufficient data on past procurements has been accumulated, even a primitive model can predict the amount of materials that the enterprise will require in the upcoming period. The forecast will be useful for purchasing department staff when planning future actions and avoiding emergency purchases, which usually cost more.
- Vendor Evaluation: At this time, vendors are selected solely on the basis of price. A simple scorecard that tracks each vendor's performance regarding timely deliveries, rejections, and/or responsiveness would give HR a much better perspective regarding each vendor's actual reliability as well as identifying which vendors are less expensive than others but may be unreliable based on their respective records.

11. Conclusion

The Procure-to-Pay cycle is much more than a sequence of SAP transactions. It determines what gets bought, from whom, at what price, and when — and getting it wrong has consequences that ripple through production schedules, cash flow, and supplier relationships.

Through this project, we walked through every stage of the P2P cycle as it operates in SAP MM — from the first Purchase Requisition raised by a department to the final payment cleared to the vendor. More importantly, we looked at the data that comes out of these transactions and tried to make sense of it — which vendors are we relying on too heavily, where is the spend going month-to-month, and where could a bit of negotiation save money.

The procurement analysis for ABC Manufacturing Ltd identified potential threats to vendor dependence, price discrepancies, and the seasonality of demand, all of which can be acted upon by the procurement manager. It is for such reasons that the knowledge of SAP processes and data analytics allows one not only to keep track of transactions but also to analyze what they mean.

It is clear that the use of SAP with analytical software like Power BI will be key to the future of procurement operations. Learning the process of procurement inside out and learning how to interpret the information from it will go a long way in helping to do that.

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